THEORETICAL ANALYSIS & ALGORITHMIC FRAMEWORK

Spectral-Laplacian Prioritized Sweeping (SL-PS)

Resolving Braess's Paradox via Resolvent Value Warm-Start & Canonical DP

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Core Tracks: Model-Based RL × Spectral Graph Theory × Congestion Games

MODEL-BASED REINFORCEMENT LEARNING • SPECTRAL GRAPH THEORY • WARDROP EQUILIBRIUM

Network Congestion Architecture, Link Latency, and Graph Laplacians

Congestion Network $G = (V, E_0)$ & Link Latency ℓ_e

- Finite directed network $G = (V, E_0)$ with $|V| = n$ vertices and $|E_0| = m$ base edges.
- Each base link $e = (u \to w) \in E_0$ carries an **affine load latency**:

$$\ell_e(f_e) = a_e f_e + b_e, \quad a_e > 0,\ b_e \geq 0$$

(a_e: congestion sensitivity; b_e: free-flow travel latency).
- Total background demand $q > 0$ from origin s to sink t, creating net injection: $p = q(e_s - e_t) \in \mathbb{R}^{|V|}$.

Graph Operators: Incidence Matrix B & Laplacian L

- **Directed Vertex–Edge Incidence Matrix** $B \in \mathbb{R}^{|V| \times |E_0|}$:

$$B_{v,u\to w} = \begin{cases} +1, & \text{if } v = u \text{ (tail)} \\ -1, & \text{if } v = w \text{ (head)} \\ 0, & \text{otherwise} \end{cases}$$

- **Conductance**: $W = \mathrm{diag}(a_e^{-1})$, **Laplacian**: $L = BWB^T \in \mathbb{R}^{|V| \times |V|}$.
- SPSD properties: $\ker(L) = \mathrm{span}(\mathbf{1})$.
- **Baseline Electrical Potential**: $\phi = L^+ p = qL^+(e_s - e_t)$.

Analytical Bridge: Matrices B and L form an exact discrete differential calculus between edge flow space ($|E_0|$) and vertex potential space ($|V|$).

The Congestion-Coupled Dynamic Graph MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{C}, \gamma)$

State Space $\mathcal{S}$ & Action Space $\mathcal{A}$

- **State Space** $\mathcal{S}$: Physical vertices $s_t = u \in V$.
- **Absorbing Destination Sink** $t_{\text{sink}} \in V$: Zero stage cost ($c(t_{\text{sink}}, \cdot) = 0$, $\mathcal{P}(t_{\text{sink}} \mid t_{\text{sink}}, \cdot) = 1$), ensuring $V^*(t_{\text{sink}}) = 0$.
- **Action Space** $\mathcal{A}(u)$: Active outgoing directed links:

$$\mathcal{A}(u) = \{v \in V \mid (u \to v) \in E_t\}$$

Unconstrained by 2D DAG grids, permitting full topological traversal across arbitrary directed graphs.

Topology Dynamics $\mathcal{P}$, Stage Cost $\mathcal{C}$ & Discount γ

- **Stochastic Topology Dynamics** $\mathcal{P}$: Shortcut chord $e^* = (u \to v)$ emerges stochastically with probability p_{chord}, yielding dynamic edge set $E_t = E_0 \cup \{e^*\}$.
- **Shortcut Link**: Nominal affine latency $\ell^*(z) = cz + d$.
- **Step Cost** $\mathcal{C}$: $c(u, v)$ couples endogenous equilibrium load $f_e^*(E_t)$ with directional detour offsets.
- **Discount Factor** $\gamma \in (0, 1)$: Governs infinite-horizon planning.

Problem Formulation: **CC-DG-MDP** formalizes a two-tier coupling: stochastic topological link emergence drives macroscopic endogenous flow re-equilibration.

Wardrop User Equilibrium (WUE) & Beckmann Optimization

Wardrop User Equilibrium (WUE)

Wardrop's First Principle (1952): Under selfish background commuter routing, all active paths $p \in \mathcal{P}_{st}$ with non-zero flow ($h_p^* > 0$) share an identical minimal latency μ_{st}:

$$\begin{cases} c_p(f^*) = \sum_{e \in p} \ell_e(f_e^*) = \mu_{st}, & \text{if } h_p^* > 0, \\ c_p(f^*) \geq \mu_{st}, & \text{if } h_p^* = 0. \end{cases}$$

No commuter can unilaterally switch routes to reduce latency. Total travel time: $C(f) = \sum_e f_e \ell_e(f_e)$.

Convex Beckmann Potential Optimization

Equilibrium flow $f^*(E_t)$ is the unique global minimizer of the strictly convex Beckmann program (1956):

$$\min_{f \geq 0} \Phi_{\text{Beck}}(f) \equiv \sum_{e \in E_t} \int_0^{f_e} \ell_e(x)\, dx \quad \text{s.t.} \quad B_{E_t} f = p$$

First-order KKT optimality conditions are mathematically identical to WUE!

Agent Flow Approximation (Scale Separation)

- Macroscopic flow f^* arises from thousands of selfish drivers; the ego-agent is an infinitesimal micro-vehicle.
- Agent precomputes nominal flow f^*_{base} **offline once**; upon chord emergence, it predicts flow shift $\Delta\vec{f}$ to evade traps.

Structural Decomposition of Step Cost Function $c(u, v)$

$$c(u, v) = \ell_e(f_e^*(E_t)) + \delta_{\text{detour}} \cdot \mathbb{I}_{\text{ret}}(u \to v) + \xi_t$$

1. Endogenous Latency $\ell_e(f_e^*)$

- Base edge $e \in E_0$:

$$\ell_e(f_e^*) = a_e f_e^*(E_t) + b_e$$

- Shortcut chord $e^* = (u \to v)$:

$$\ell^*(z^*) = cz^* + d$$

- As $z^* \to 0$, shortcut nominal latency appears cost-free ($c \approx 0$).
- Physical law: Swarming traffic causes downstream delays to surge!

2. Retrograde Penalty $\delta_{\text{det}}\mathbb{I}_{\text{ret}}$

- Let $d(x) \equiv \text{dist}_G(x, t_{\text{sink}})$:

$$\mathbb{I}_{\text{ret}} = \begin{cases} 1, & d(v) \geq d(u) \\ 0, & d(v) < d(u) \end{cases}$$

- Eliminates zero-gain dithering oscillations on reversible networks.
- **Key property**: Braess chords advance forward ($d(v) < d(u)$), so $\mathbb{I}_{\text{ret}} = 0$; penalty never masks shortcut appeal!

3. Observation Noise & Sink

- **Stochastic Noise** ξ_t:

$$\xi_t \sim \mathcal{N}(0, \sigma_\xi^2)$$

Models real-time sensor measurement errors.

- **Destination Absorption** t_{sink}:

$$c(t_{\text{sink}}, a) = 0, \quad \forall a$$

Ensures true optimal value $V^*(t_{\text{sink}}) = 0$.

Braess's Paradox in Congestion-Coupled Graphs

Nature of Braess's Paradox

- **Paradox**: Opening a new shortcut link $e^* = (u \to v)$ with nominal cost $c \approx 0$ paradoxically **increases total travel time**:

$$C(f^*_{\text{aug}}) > C(f^*_{\text{base}})$$

- **Mechanism**: Selfish drivers swarm into the shortcut, causing severe bottlenecks downstream of exit vertex v.
- **Failure of Standard-PS**: Deceived by local nominal cost $c \approx 0$, it plunges into the trap; congestion takes $\mathcal{O}(\text{diam}(G))$ updates to propagate upstream.

Generalized Voltage Criterion (Theorem 1)

- Shortcut diverts flow ($z^* > 0$) if and only if:

$$\Delta b_{uv} + V_{uv} > d$$

where $V_{uv} = \phi_u - \phi_v = q(e_s - e_t)^T L^+ (e_u - e_v)$.

- **Braess's paradox occurs whenever flow opposes electrical gradient**:

$$V_{uv} = \phi_u - \phi_v < 0$$

- Network delay increase: $\Delta C \leq |V_{uv}| z^* > 0$.
- On grids $G_{m,n}$, $\Delta b_{uv} = k \cdot b$ ($k > 0$) and $V_{uv} < 0$.

Consequence: Standard RL planners greedily enter the deceptive shortcut and incur catastrophic cumulative delay $\Omega(q)$.

Core Components of SL-PS: $\Delta\vec{f}$, $\Delta\vec{\ell}$, and Divergence ξ_{div}

1. Base Edge Flow Shift $\Delta\vec{f} \in \mathbb{R}^{|E_0|}$

$$\Delta\vec{f} = f^*_{\text{aug},E_0} - f^*_{\text{base}}$$

- $f^*_{\text{base}} \in \mathbb{R}^{|E_0|}$: Base equilibrium flow on original network E_0.
- f^*_{aug,E_0}: Re-equilibrated flow on E_0 after shortcut chord e^* opens.
- **Meaning**: Relieves load on alternate paths ($\Delta f_e < 0$) while overloading downstream bottleneck links ($\Delta f_e > 0$).

2. Link Latency Shift $\Delta\vec{\ell} \in \mathbb{R}^{|E_0|}$

$$\Delta\vec{\ell} = \text{diag}(a_e)\Delta\vec{f}$$

Since $\ell_e(f_e) = a_e f_e + b_e$, link derivative is a_e. Latency shift is $\Delta\ell_e = a_e \Delta f_e$ (seconds).

3. Vertex Congestion Divergence $\xi_{\text{div}} \in \mathbb{R}^{|V|}$

Projects link latency shifts onto vertices via incidence matrix B:

$$\xi_{\text{div}} = |B\Delta\vec{\ell}| \in \mathbb{R}^{|V|}$$

Explicit expansion at each vertex $v \in V$:

$$\xi_{\text{div}}(v) = \left| \sum_{v \to w} a_e \Delta f_e - \sum_{u \to v} a_e \Delta f_e \right|$$

Physical Interpretation:

- B serves as a **discrete divergence operator**, measuring local net latency flux imbalance.
- At shortcut exit v^*, massive convergent traffic creates a divergence shock: $\xi_{\text{div}}(v^*) = \Omega(q)$.
- Absolute value $|\cdot|$ provides positive source excitation for the spectral potential.

Flow Kernel P_{flow}, Green Operator $\mathcal{G}_{\text{out}}$, and Potential Φ_{spec}

Flow-Preserving Markov Kernel P_{flow}

Transition probabilities follow actual commuter choices:

$$P^{\text{flow}}_{u,v} = \frac{f^*_{\text{base},(u\to v)}}{\sum_w f^*_{\text{base},(u\to w)}}, \quad P^{\text{flow}}_{t_{\text{sink}},t_{\text{sink}}} = 1$$

Completely eliminates branching over-diffusion on asymmetric graph topologies.

Green Operator $\mathcal{G}_{\text{out}}$ & Successor Representation

Out-Laplacian resolvent: $\mathcal{G}_{\text{out}} = (I - P + \eta I)^{-1} = \sum_{k=0}^{\infty} \frac{P^k}{(1+\eta)^{k+1}}$.
Setting $\eta = \frac{1-\gamma}{\gamma}$ yields the exact algebraic identity:

$$\mathcal{G}_{\text{out}} \equiv \gamma(I - \gamma P)^{-1} \equiv \gamma \mathcal{R}_\gamma \quad \text{(SR Resolvent)}$$

Spectral Congestion Potential Φ_{spec}

$$\Phi_{\text{spec}} \equiv \frac{1}{\gamma}\mathcal{G}_{\text{out}}\xi_{\text{div}} = (I - \gamma P)^{-1}\xi_{\text{div}} = \sum_{k=0}^{\infty} \gamma^k P^k \xi_{\text{div}}$$

- **Infinite-Horizon Projection**: Linear solve $(I - \gamma P)^{-1}\xi_{\text{div}}$ projects downstream delays upstream in $\mathcal{O}(1)$ time.
- **Dimensionally Consistent**: Measured in seconds of delay.

Spectral Resolvent Value Warm-Start

$$V^{(0)}(s) = V^*_{\text{base}}(s) + \beta(\sigma)\Phi_{\text{spec}}(s)$$

with $\beta(\sigma) = \beta_0(1 + \sigma(P))$. Instantly elevates bottleneck value $V^{(0)} \implies$ **Deterministic trap immunity at zero budget ($K = 0$)!**

Comprehensive SL-PS Algorithmic Workflow (Algorithm 1)

Algorithm 1 Spectral-Laplacian Prioritized Sweeping (SL-PS)

Require: General base network $G = (V, E_0)$, demand q, budget K, discount γ, regularizer $\eta = \frac{1-\gamma}{\gamma}$, scale β_0.

Offline: Compute L, L^+, base potentials $\phi = qL^+(e_s - e_t)$, base equilibrium f^*_{base}, base value function V^*_{base}, flow-preserving transition matrix $P = P_{\text{flow}}$ (Eq. (19)), SLEM $\sigma(P)$, and Green operators $\mathcal{G}_{\text{out}} = (I - P + \eta I)^{-1}$, $\mathcal{G}_{\text{in}} = (D_{\text{in}} - P^T + \eta I)^{-1}$.
Initialize $V \leftarrow V^*_{\text{base}}$, Priority Queue $Q \leftarrow \emptyset$.
for navigation step $t = 1, 2, \dots$ until reaching sink t_{sink} **do**
  Observe network link status.
  if candidate shortcut $e^* = (u \to v)$ has newly emerged at step t **then**
    Compute base edge flow shift $\Delta\vec{f}$ via closed-form electrical shift (Eq. (18)) or augmented equilibrium solver.
    Compute node divergence $\xi_{\text{div}} = |B(a\Delta\vec{f})|$ and spectral resolvent potential $\Phi_{\text{spec}} = \frac{1}{\gamma}\mathcal{G}_{\text{out}}\xi_{\text{div}}$.
    Modulate spectral gain $\beta \leftarrow \beta_0(1 + \sigma(P))$.
    Spectral Resolvent Value Warm-Start: $V(s) \leftarrow V^*_{\text{base}}(s) + \beta\Phi_{\text{spec}}(s)$ for all $s \in V$.
  else if active shortcut e^* has vanished **then**
    $V(s) \leftarrow V^*_{\text{base}}(s)$, $\Phi_{\text{spec}} \leftarrow \mathbf{0}$, $\beta \leftarrow 0$.
  end if
  Compute canonical Bellman residual vector $\xi_{\text{res}}(s) = |TV(s) - V(s)|$.
  Backward potential attribution: $f_{\text{in}} = \mathcal{G}_{\text{in}}(\xi_{\text{res}} + \xi_{\text{div}})$.
  for each state $s \in V$ **do**
    Priority key: $\text{prio}(s) \leftarrow \xi_{\text{res}}(s) + \beta f_{\text{in}}(s)$.
    if $\text{prio}(s) > \varepsilon$ **then** Q.insert$(s, \text{prio}(s))$.
  end for
  for backup iteration $k = 1, \dots, K$ **do**
    if Q is empty **then break**
    $s \leftarrow Q$.pop_max().
    $V(s) \leftarrow TV(s)$ {Canonical Bellman backup, Eq. (26)}
    for each predecessor $p \in \text{Pred}(s)$ **do**
      Update Q.insert$(p, |TV(p) - V(p)| + \beta f_{\text{in}}(p))$.
    end for
  end for
  Execute canonical greedy action $v^* = \text{argmin}_{w \in \mathcal{A}(s_t)}[c(s_t, v) + \gamma V(v)]$.
  Transition to adjacent vertex $s_{t+1} = v^*$ and observe step latency.
end for

Phase 1: Offline Precomputation

- Compute L, L^+, potential $\phi = qL^+(e_s - e_t)$, base flow f^*_{base}, base value V^*_{base}.
- Construct P_{flow}, SLEM $\sigma(P)$, and Green operators $\mathcal{G}_{\text{out}}$, $\mathcal{G}_{\text{in}}$.
- Heavy matrix operations are solved entirely offline!

Phase 2: Instantaneous Warm-Start ($K = 0$ Shield)

- Upon chord emergence e^*: compute $\Delta\vec{f}$, $\xi_{\text{div}} = |B(a\Delta\vec{f})|$, and Φ_{spec}.
- Warm-start value function: $V^{(0)} \leftarrow V^*_{\text{base}} + \beta(\sigma)\Phi_{\text{spec}}$.
- **100% Trap Immunity**: Bottleneck state value is elevated truthfully $\implies$ Agent circumvents trap with zero sweeping sweeps!

Phase 3: Canonical Swept Backups ($K \geq 1$ Optimizer)

- Backward Green potential scheduling: $f_{\text{in}} = \mathcal{G}_{\text{in}}(\xi_{\text{res}} + \xi_{\text{div}})$.
- Prioritizes upstream branching junctions, eliminating queue starvation.
- Executes canonical Bellman backups $\mathcal{T}V$ to close detour gap toward Pareto optimality.

Theoretical Foundations & Key Theorems Suite

Theorem 2: Strict Contraction Mapping

$\|\mathcal{T}V_1 - \mathcal{T}V_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$. Canonical Bellman operator guarantees global convergence to the true physical optimum V^*, uncorrupted by heuristic penalties.

Theorem 3: Tight Immunity & Vanishing Threshold

Greedy policy at $K = 0$ avoids Braess trap if $\beta_0 \geq \beta^*_{\min}(u) = \frac{\Delta Q_{\text{base}}(u)}{\gamma(1+\sigma(P))\Delta\Phi_{\text{spec}}(u)}$. Under heavy traffic ($q \to \infty$), $\Delta\Phi_{\text{spec}} = \Omega(q)$ forces $\beta^*_{\min} \to 0$! Any nominal $\beta_0 = 1$ is unconditionally immune.

Theorem 4: Pareto-Optimal Convergence

Value error contracts geometrically after K sweeps: $\|V^{(K)} - V^*\|_\infty \leq \gamma^K \epsilon \to 0$. Rapidly closes detour regret gap.

Theorem 5: Asymptotic Policy Divergence

Cumulative travel latency gap between Standard-PS and SL-PS diverges: $\mathbb{E}[T_{\text{PS}} - T_{\text{SL-PS}}] = \Omega(q \cdot \text{diam}(G)) \to \infty$. Canonical PS fails catastrophically on large networks.

Theorem 6: Lipschitz Fault Tolerance

Under flow estimation errors $\|\hat{f} - f^*\| \leq \epsilon_f$: $\|V^{(0)} - \hat{V}^{(0)}\|_\infty \leq \mathcal{L}_{\text{SL}}\epsilon_f$. Strictly positive tolerance margin $\epsilon^*_f > 0$ preserves zero-trap immunity.

Theorem 7: Policy–Resolvent Mismatch

Resolvent Identity II: $\|\Phi(P) - \Phi(P_{\pi^*})\|_\infty \leq \frac{\gamma}{(1-\gamma)^2}\|\xi\|_\infty \epsilon_P$. Flow kernel P_{flow} eliminates error; unweighted P_{unif} remains provably safe for $\epsilon_P < \epsilon^*_P$.

Core Takeaway: SL-PS reconciles **Instantaneous Safety** at $K = 0$ with **Canonical Optimality** at $K \geq 1$ across general arbitrary networks.